

Quantum Mechanics Griffith - Solutions Manual

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Contents

1 The Wave Function	5
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Chapter 1

The Wave Function

Probability

Problem 1.1 - 1.2

Too basic. Learn probability if cannot solve.

Problem 1.3

(a) Since it's uniform, we want area of rectangle $\pi\rho = 1 \Rightarrow \rho = \frac{1}{\pi}$ for $\theta \in [0, \pi]$.
Obviously, $\rho = 0$ for $\theta \in [-\frac{\pi}{2}, 0) \cup (\pi, \frac{3\pi}{2}]$.

(b)

$$\langle \theta \rangle = \frac{\pi}{2}$$

$$\langle \theta^2 \rangle = \int_0^\pi \theta^2 \rho(\theta) d\theta = \int_0^\pi \theta^2 \frac{1}{\pi} d\theta = \frac{1}{3\pi} [\theta^3]_0^\pi = \frac{\pi^2}{3}$$

(c)

$$\langle \sin \theta \rangle = \int_0^\pi \sin \theta \frac{1}{\pi} d\theta = -\frac{1}{\pi} [\cos \theta]_0^\pi = \frac{2}{\pi}$$

Cosine is odd around $\theta = \frac{\pi}{2}$. So $\langle \cos \theta \rangle = 0$.

$$\langle \cos^2 \theta \rangle = \int_0^\pi \cos^2 \theta \frac{1}{\pi} d\theta = \frac{1}{\pi} \left[\frac{\theta}{2} + \frac{\sin 2\theta}{4} \right]_0^\pi = \frac{1}{2}$$

Problem 1.4

$$\rho(\theta) = \frac{1}{\pi}$$

$$F(\theta) = \frac{1}{\pi} \theta$$